



Electronics trainer kits at an incubator facility desk

industry. These are helping students learn, innovate and design latest embedded projects at a quicker pace. Latest technology kits are provided with step-by-step procedure manuals, tutorial videos or animations, for easy and clear usability.

This is a major aid towards the development of skills as per industry requirement. Simultaneously, these new technologies keep the excitement among the students alive, which, in turn, brings a positive impact on the demand for these products.

India's market and ecosystem

The education and training sector in India is quite strong and drives a significant percentage of the economy. A report by India Brand Equity Foundation (IBEF), Ministry of Commerce and Industry, government of India, mentions that, India's education sector is valued at US\$ 91.7 billion, as of April 2018. It is projected to touch US\$ 180 billion by 2020. Demand of training kits and products will proportionally increase with the increasing number of educational institutes providing higher education and practical training, as well as with the strengthening of technology industries.

The training products market saw a significant rise in business between 2008 and 2012, due to the increasing number of engineering and polytechnics institutions. That pace took to stagnancy afterwards, when admission counts fell, especially in electronics and telecommunication engineering. However, demand is still significant for these training kits, as reputed institutes receive enough applications for these branches and keep upgrading their laboratories with

modern devices consistently.

Dilip Jain, chief executive officer, Rajguru Electronics, is optimistic. He says, "While data suggests people are not preferring electronics as their first choice of education, raising concerns regarding the demand, the reality came as a contradiction—demand is increasing every year. Many new domestic suppliers are entering the market and the competition is quite high."

With 850 universities as of April 2018, with plans to raise ₹ 200 billion for setting up six new IITs by 2024, demand for hands-on training products will increase. Rising number of incubation centres and Atal Tinkering Labs also stand evidence to this fact.

Experts unanimously agree that demand for sensors, the IoT and robotic kits is reaching new heights. Sagar Gala, director, Gala Electronics, says, "There is a good balance between demand and supply in the kits industry. Domestic manufacturers are also playing a big role in the education industry."

But is domestic production growing at a satisfactory scale? Agarwal answers, "Most training kit providers in the country are importers. There are very few indigenous products."

By 2019, the Indian education industry is expected to become a US\$ 101.1 billion market. It will be interesting to see new kits/companies coming up in the next couple of years, given the predictions in place.

Market challenges

The Indian market still suffers from many obstacles. These have impacted end products in terms of cost, stock or availability, quality and so on.

As per industry players, significant obstacles for the electronics educational kits market include high tax rates (13.5% VAT being increased to 18% GST), deviation in demand due to recession or lack of admissions in a specific engineering branch, stagnant and redundant government policies, financial stress on institutes and slow upgrade to new technologies.

As a possible solution, they advise the GST to be exempted, looking at

the positive role trainer kits play in the education system. Also, basic electronics and skill-based subjects should have more focus and must be taught more thoroughly in schools. This will pique the interest of students, mold their practical skills early and, consequently, benefit the market and the industry.

Another major challenge pointed out by the industry is the inefficient local ecosystem for manufacturing kits. Agarwal adds, "The fact is, standalone educational kits are not the complete solution. Training programmes should define what these require from kits to meet learning outcomes."

"There is also a need to evangelise the market. For training programmes, the major challenge is lack of industry-academia partnerships, which need to be looked into immediately."

Moreover, understanding and awareness among customers is missing. This has led to a trust deficit. Experts say that people in India do not easily trust Indian companies. While quite a few international brands enjoy big market shares despite providing ridiculously expensive products, that too based on old technology, India-based manufacturers struggle to create a strong footprint.

Apart from that, generic Arduino-based kits from Chinese websites (like AliExpress, BangGood and Club Factory) have started flooding the market. These are available at extremely low prices, creating a stiffer competition for domestic players. Yet, Indian companies are producing tailor-made learning kits as per market demands. This awareness needs to be spread.

National institutions like the IITs and NITs should take the onus to source more kits from domestic players to support Make in India policy of the government, the experts suggest. Velanti says, "Some IITs are still importing training kits. This needs to be addressed."

"Additionally, the government of India should give subsidies to GESS and Worlddidac exhibitions, to enhance quality and exposure of Indian products, and market these all over the world to improve exports."

Dr Raut asserts, "Training kit manufacturers, including us, run with